Lecture 16: Difference-in-differences

Economics 326 — Introduction to Econometrics II

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Motivation

- Previous lecture: treatment effects from **cross-sectional** data, assuming selection on observables (treatment is as good as random after controlling for covariates).
- Often hard to justify. Alternative: exploit **panel data** (repeated observations on the same units over time).
- **Difference-in-differences (DID)**: compare changes over time between a treatment group and a control group.

DID basic setup

- Two time periods: $t \in \{0, 1\}$ (before and after treatment).
- Two groups: $D_i \in \{0, 1\}$ (control and treatment).
- Treatment occurs between periods 0 and 1; only the treatment group ($D_i = 1$) is affected.
- Y_{it}: outcome for individual i at time t.

DID regression model

- DID regression:

$$Y_{it} = \alpha + \delta \cdot t + \gamma D_i + \beta(t \cdot D_i) + U_{it},$$

  where $\mathrm{E}\left[U_{it} \mid D_i\right] = 0$.
- Regressors:
  - t: time (0 = before, 1 = after).
  - D_i: group (0 = control, 1 = treatment).
  - $t \cdot D_i$: interaction, equals 1 only for the treatment group after treatment.

Interpreting the coefficients

- $$Y_{it} = \alpha + \delta \cdot t + \gamma D_i + \beta(t \cdot D_i) + U_{it},$$
- Evaluate $\mathrm{E}\left[Y_{it} \mid D_i\right]$ for each combination of t and D_i:

$$\begin{aligned}
t = 0,\ D_i = 0: &\quad \mathrm{E}\left[Y_{i0} \mid D_i = 0\right] = \alpha, \\
t = 0,\ D_i = 1: &\quad \mathrm{E}\left[Y_{i0} \mid D_i = 1\right] = \alpha + \gamma, \\
t = 1,\ D_i = 0: &\quad \mathrm{E}\left[Y_{i1} \mid D_i = 0\right] = \alpha + \delta, \\
t = 1,\ D_i = 1: &\quad \mathrm{E}\left[Y_{i1} \mid D_i = 1\right] = \alpha + \delta + \gamma + \beta.
\end{aligned}$$

- Summarized as a 2×2 table:

	$D_i = 0$ (Control)	$D_i = 1$ (Treatment)
$t = 0$	α	$\alpha + \gamma$
$t = 1$	$\alpha + \delta$	$\alpha + \delta + \gamma + \beta$

- α: baseline (control group, $t = 0$).
- γ: pre-existing **group difference** at $t = 0$.
- δ: **time effect** — change in the control group from $t = 0$ to $t = 1$ (common trend).
- Change over time for each group:
  - **Treatment:** $\mathrm{E}[Y_{i1} \mid D_i = 1] - \mathrm{E}[Y_{i0} \mid D_i = 1] = \delta + \beta$.
  - **Control:** $\mathrm{E}[Y_{i1} \mid D_i = 0] - \mathrm{E}[Y_{i0} \mid D_i = 0] = \delta$.
- Subtract control's change from treatment's change:

$$(\delta + \beta) - \delta = \beta.$$

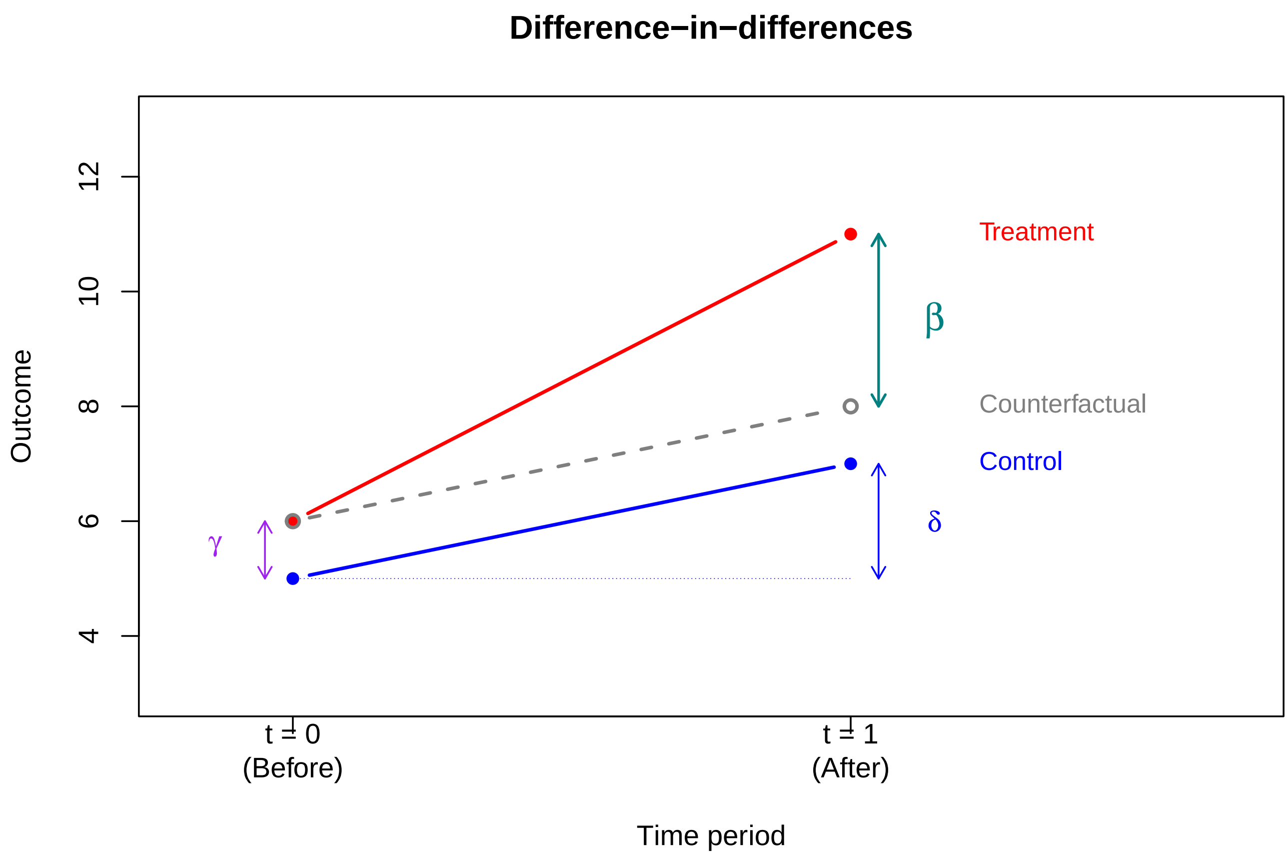
- DID estimand as a double difference:

$$\beta = \mathrm{E}[Y_{i1} - Y_{i0} \mid D_i = 1] - \mathrm{E}[Y_{i1} - Y_{i0} \mid D_i = 0].$$

- The common trend δ cancels, isolating β.

DID diagram

- DID diagram — control, treatment, and counterfactual:



- Counterfactual (dashed gray): treatment group's baseline $\alpha + \gamma$ plus the control group's change δ.

DID and potential outcomes

- **Panel potential outcomes**: $Y_{it}(d)$ — outcome for individual i at time t if assigned to group $d \in \{0, 1\}$.
- Four potential outcomes: $Y_{i0}(0)$, $Y_{i1}(0)$, $Y_{i0}(1)$, $Y_{i1}(1)$.
- The observed outcome is:

$$Y_{it} = D_i\, Y_{it}(1) + (1 - D_i)\, Y_{it}(0).$$

- What we observe for each group:

	Control ($D_i = 0$)	Treatment ($D_i = 1$)
$t = 0$	$Y_{i0}(0)$	$Y_{i0}(1)$
$t = 1$	$Y_{i1}(0)$	$Y_{i1}(1)$

- Treatment effect on the treated at $t = 1$:

$$\text{ATT} = \text{E}\left[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1\right].$$

  The counterfactual $Y_{i1}(0)$ is unobserved for the treated group.

DID as a treatment effect

- $\beta = \text{E}\left[Y_{i1} - Y_{i0} \mid D_i = 1\right] - \text{E}\left[Y_{i1} - Y_{i0} \mid D_i = 0\right]$.
- Substituting observed outcomes with potential outcomes:

$$\begin{aligned}\beta = {} & \text{E}\left[Y_{i1}(1) - Y_{i0}(1) \mid D_i = 1\right] \\ & - \text{E}\left[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0\right].\end{aligned}$$

- To relate β to the ATT, add and subtract $Y_{i1}(0)$ and $Y_{i0}(0)$ inside the first expectation:

$$\begin{aligned}\beta &= \text{E}\left[Y_{i1}(1) - Y_{i0}(1) \mid D_i = 1\right] - \text{E}\left[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0\right] \\ &= \text{E}\left[Y_{i1}(1) \underbrace{- Y_{i1}(0) + Y_{i1}(0)}_{=0} \underbrace{- Y_{i0}(0) + Y_{i0}(0)}_{=0} - Y_{i0}(1) \mid D_i = 1\right] \\ &\quad - \text{E}\left[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0\right].\end{aligned}$$

- Rearranging and splitting the expectation:

$$\begin{aligned}\beta = {} & \underbrace{\text{E}\left[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1\right]}_{\text{ATT}} \\ & + \underbrace{\text{E}\left[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1\right] - \text{E}\left[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0\right]}_{\text{difference in trends}} \\ & + \underbrace{\text{E}\left[Y_{i0}(0) - Y_{i0}(1) \mid D_i = 1\right]}_{\text{anticipation effect}}.\end{aligned}$$

- For β to equal the ATT, the difference in trends and anticipation effect must be zero. This requires two assumptions.

Assumption 1: Parallel trends

- Difference in trends is zero if both groups would have experienced the same change over time absent treatment:

$$\mathrm{E}\left[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1\right] = \mathrm{E}\left[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0\right].$$

- Under parallel trends, the decomposition reduces to:

$$\beta = \mathrm{ATT} + \underbrace{\mathrm{E}\left[Y_{i0}(0) - Y_{i0}(1) \mid D_i = 1\right]}_{\text{anticipation effect}}.$$

- Cannot be directly tested: $Y_{i1}(0)$ is unobserved for the treated group.
- With multiple pre-treatment periods, can check whether trends were parallel before treatment.

Assumption 2: No anticipation

- Anticipation effect is zero if pre-treatment outcomes are not affected by future treatment assignment:

$$\mathrm{E}\left[Y_{i0}(1) \mid D_i = 1\right] = \mathrm{E}\left[Y_{i0}(0) \mid D_i = 1\right].$$

- Treatment assignment does not change pre-treatment outcomes in expectation.
- Under both parallel trends and no anticipation:

$$\beta = \mathrm{ATT}.$$

Example: incinerator and house prices

- Kiel and McClain (1995): did a garbage incinerator in North Andover, MA reduce nearby house prices?
- Data: `kielmc` from the `wooldridge` package:

```r
library(wooldridge)
data(kielmc)
# Show 2 observations from each of the 4 groups
rows <- c(
  head(which(kielmc$y81 == 0 & kielmc$nearinc == 0), 2),
  head(which(kielmc$y81 == 0 & kielmc$nearinc == 1), 2),
  head(which(kielmc$y81 == 1 & kielmc$nearinc == 0), 2),
  head(which(kielmc$y81 == 1 & kielmc$nearinc == 1), 2)
)
kielmc[rows, c("rprice", "y81", "nearinc", "age")]
```

```
       rprice y81 nearinc age
14   52000.00   0       0  32
15   49000.00   0       0  18
1    60000.00   0       1  48
2    40000.00   0       1  83
187  90245.77   1       0   1
188  46082.95   1       0  41
180  37634.41   1       1  81
181  39938.55   1       1  71
```

- `rprice`: house price in 1978 dollars (Y_{it}).

- y81: 1 if year is 1981 (after incinerator announced), 0 if 1978 ($t = 1$ if 1981, 0 if 1978).
- nearinc: 1 if house is near the incinerator site (D_i).

The 2×2 table of means

- Compute the four group means:

```r
means <- tapply(kielmc$rprice, list(kielmc$y81, kielmc$nearinc), mean)
colnames(means) <- c("Far (nearinc=0)", "Near (nearinc=1)")
rownames(means) <- c("1978 (y81=0)", "1981 (y81=1)")
round(means, 2)
```

```
             Far (nearinc=0) Near (nearinc=1)
1978 (y81=0)        82517.23         63692.86
1981 (y81=1)       101307.51         70619.24
```

- Computing the DID by hand:

```r
diff_near <- means[2, 2] - means[1, 2]
diff_far  <- means[2, 1] - means[1, 1]
DID <- diff_near - diff_far
cat("Change (near):", round(diff_near, 2), "\n")
```

```
Change (near): 6926.38
```

```r
cat("Change (far): ", round(diff_far, 2), "\n")
```

```
Change (far):  18790.29
```

```r
cat("DID:          ", round(DID, 2), "\n")
```

```
DID:           -11863.9
```

DID regression

- The DID regression, where y81nrinc = y81 × nearinc is the interaction term:

```r
options(scipen = 999)
reg_did <- lm(rprice ~ y81 + nearinc + y81nrinc, data = kielmc)
round(summary(reg_did)$coefficients, 4)
```

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  82517.23   2726.910 30.2603   0.0000
y81          18790.29   4050.065  4.6395   0.0000
nearinc     -18824.37   4875.322 -3.8612   0.0001
y81nrinc    -11863.90   7456.646 -1.5911   0.1126
```

- Coefficient on y81nrinc matches the DID from the 2×2 table.
- $\hat{\beta} = -\$11{,}864$ (SE $= 7{,}457$, $p = 0.113$): negative but not significant at 5%.

Assumptions in the incinerator example

- **Parallel trends:** without the incinerator, house prices near and far from the site would have followed the same trend over time.
- **No anticipation:** before the incinerator was announced, living near the future site did not affect house prices.

DID with covariates

- Basic DID requires parallel trends to hold unconditionally. If near houses are systematically different (e.g., older) and age affects price trends, the groups may follow different trajectories even without the incinerator.

- Omitting age biases δ, which biases β.
- Adding covariates: **parallel trends conditional on age**. Among houses of the same age, near and far would follow the same trend.

```r
reg_did_cov <- lm(rprice ~ y81 + nearinc + y81nrinc + age + I(age^2),
                  data = kielmc)
round(summary(reg_did_cov)$coefficients, 4)
```

```
               Estimate Std. Error  t value Pr(>|t|)
(Intercept)  89116.5354  2406.0511  37.0385   0.0000
y81          21321.0418  3443.6311   6.1914   0.0000
nearinc       9397.9359  4812.2218   1.9529   0.0517
y81nrinc    -21920.2700  6359.7454  -3.4467   0.0006
age          -1494.4240   131.8603 -11.3334   0.0000
I(age^2)         8.6913     0.8481  10.2476   0.0000
```

- With age controls: $\hat{\beta} = -\$21{,}920$ (SE = 6,360, $p = 0.0006$) — nearly twice as large, highly significant.
- The change suggests the basic DID was biased by age-driven trend differences.

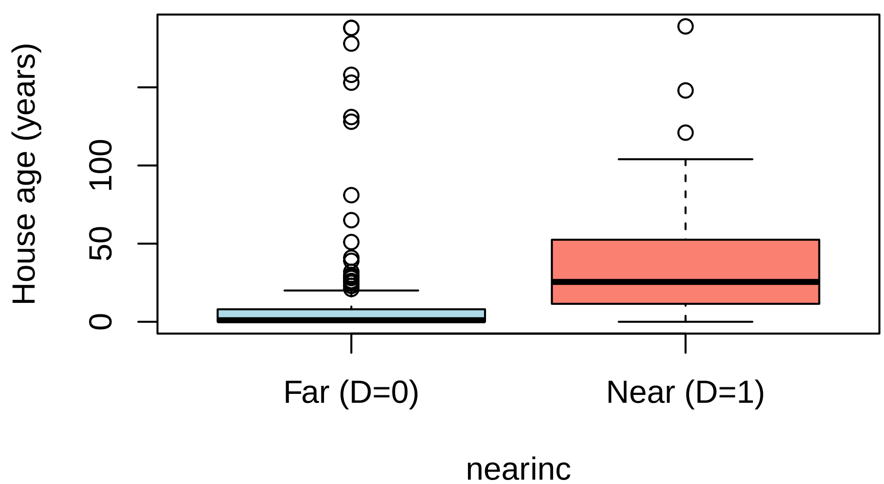
Where does the bias come from?

- Under no anticipation, the decomposition from the earlier slide gives:

$$\begin{aligned}\beta &= \text{ATT} \\ &\quad + \underbrace{\mathrm{E}\left[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1\right] - \mathrm{E}\left[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0\right]}_{\text{difference in trends}}.\end{aligned}$$

- **No anticipation is plausible:** incinerator not publicly known in 1978.
- Bias must come from difference in trends: a **parallel trends** violation.
- Near houses are systematically older:

```r
boxplot(age ~ nearinc, data = kielmc,
        names = c("Far (D=0)", "Near (D=1)"),
        ylab = "House age (years)",
        col = c("lightblue", "salmon"))
```



- Age affects price trends nonlinearly, so near and far groups would have had **different price trends even without the incinerator**.
- Estimated bias: $-\$11{,}864 - (-\$21{,}920) \approx +\$10{,}056$ — the age-driven trend difference partially masked the incinerator's negative effect.

Assumption	Violated?	Reasoning
No anticipation	No	Incinerator not yet announced in 1978
Parallel trends	Yes (unconditionally)	Near houses are older; older houses had different price trends

Detecting anticipation and pre-trends

- With only two time periods, anticipation effects **cannot** be separately identified.
- With **multiple pre-treatment periods**, use an **event study** design. Data span $t = -T, \ldots, -1, 0, 1, \ldots, T'$, treatment at $t = 0$. Replace the single interaction $\beta(t \cdot D_i)$ with a separate coefficient per period:

$$Y_{it} = \alpha + \sum_{s \neq 0} \delta_s \cdot \mathbb{1}[t = s] + \gamma D_i + \sum_{s \neq 0} \beta_s \cdot D_i \cdot \mathbb{1}[t = s] + U_{it}.$$

- δ_s: time effect in period s relative to $t = 0$ (generalizes δ). β_s: treatment-control difference in period s, beyond γ and δ_s.
- The pre-treatment coefficients β_s for $s < 0$ serve as a diagnostic:

Pattern in β_s for $s < 0$	Interpretation
All close to zero: $\beta_{-3} \approx \beta_{-2} \approx \beta_{-1} \approx 0$	Both assumptions look plausible
Sharp break right before treatment: $\beta_{-3} \approx \beta_{-2} \approx 0$ but $\beta_{-1} \neq 0$	**Anticipation**: units react to news of upcoming treatment
Gradual trend across pre-periods: $\lvert\beta_{-3}\rvert < \lvert\beta_{-2}\rvert < \lvert\beta_{-1}\rvert$, growing steadily	**Parallel trends violation**, not anticipation: groups were already diverging before treatment

Time fixed effects

- In the two-period model, the term $\delta \cdot t$ creates two intercepts: α at $t = 0$ and $\alpha + \delta$ at $t = 1$.
- With multiple periods $t = 0, 1, 2, \ldots$, we cannot use $\delta \cdot t$ because that forces a **linear** trend. Instead, include a separate dummy for each period. The regression actually estimated is:

$$Y_{it} = \alpha + \delta_1 \cdot \mathbb{1}[t = 1] + \delta_2 \cdot \mathbb{1}[t = 2] + \cdots + \gamma D_i + \cdots + U_{it}.$$

- The key property: $\mathbb{1}[t = s]$ equals 1 when $t = s$ and 0 otherwise. At $t = 0$ all dummies are zero (baseline). At any other t, exactly one dummy equals 1:

Period	Active dummy	Intercept
$t = 0$	none (baseline)	α
$t = 1$	$\mathbb{1}[t = 1] = 1$	$\alpha + \delta_1$
$t = 2$	$\mathbb{1}[t = 2] = 1$	$\alpha + \delta_2$

- Each period gets its own intercept. The compact notation $\sum_{s \neq 0} \delta_s \cdot \mathbb{1}[t = s]$ writes the time dummies as a sum. The event study model with all the dummies written out:

$$
\begin{aligned}
Y_{it} = \alpha + \delta_1 \mathbb{1}[t{=}1] + \delta_2 \mathbb{1}[t{=}2] + \cdots \\
+ \gamma D_i + \beta_1 D_i \mathbb{1}[t{=}1] + \beta_2 D_i \mathbb{1}[t{=}2] + \cdots + U_{it}.
\end{aligned}
$$

- At each t, exactly one time dummy survives (and at $t = 0$ none do). So $\alpha + \delta_t$ is the intercept at time t. Define $\lambda_t = \alpha + \delta_t$ (with $\delta_0 = 0$):

$$\lambda_0 = \alpha, \quad \lambda_1 = \alpha + \delta_1, \quad \lambda_2 = \alpha + \delta_2, \quad \ldots$$

The λ_t's are called **time fixed effects**. This is the regression we actually run in OLS, with a dummy for each period:

$$
\begin{aligned}
Y_{it} = \lambda_0 \mathbb{1}[t{=}0] + \lambda_1 \mathbb{1}[t{=}1] + \lambda_2 \mathbb{1}[t{=}2] + \cdots \\
+ \gamma D_i + \beta_1 D_i \mathbb{1}[t{=}1] + \cdots + U_{it}.
\end{aligned}
$$

- At each t, exactly one λ-dummy equals 1 and all others are zero:

$$\underbrace{\lambda_0 \cdot 0 + \cdots + \lambda_t \cdot 1 + \cdots}_{\text{only } \lambda_t \text{ survives}} = \lambda_t.$$

The notation λ_t is shorthand for this — it represents whichever λ is active at time t:

$$Y_{it} = \lambda_t + \gamma D_i + \sum_{s \neq 0} \beta_s \cdot D_i \cdot \mathbb{1}[t = s] + U_{it}.$$

Individual fixed effects

- The same idea can be applied to individuals. In the event study model, the intercept for individual i is $\alpha + \gamma D_i$. This allows only **two** values:

Group	Intercept
Control ($D_i = 0$)	α
Treatment ($D_i = 1$)	$\alpha + \gamma$

All individuals within the same group share the same baseline. But individuals may differ even within a group (e.g., houses near the incinerator differ in size, age, neighborhood quality).

- **Individual fixed effects** give each individual its own dummy, just like time fixed effects give each period its own dummy. The regression is:

$$
\begin{aligned}
Y_{it} = \alpha_1 \mathbb{1}[i{=}1] + \cdots + \alpha_n \mathbb{1}[i{=}n] \\
+ (\text{time and treatment terms}) + U_{it}.
\end{aligned}
$$

Only the dummy for individual i is active, so the intercept is α_i. Using time fixed effects λ_t from the previous slide:

$$
\begin{aligned}
Y_{it} = \alpha_i + \lambda_t + \cancel{\gamma D_i} \\
+ \sum_{s \neq 0} \beta_s \cdot D_i \cdot \mathbb{1}[t = s] + U_{it}.
\end{aligned}
$$

- Since D_i does **not change over time** for any individual, it is already captured by α_i. Including both α_i and γD_i would be **perfect multicollinearity**, so we drop γD_i:

$$\begin{aligned} Y_{it} = {} & \alpha_i + \lambda_t \\ & + \sum_{s \neq 0} \beta_s \cdot D_i \cdot \mathbb{1}[t = s] + U_{it}. \end{aligned}$$

- The interaction terms $\beta_s \cdot D_i \cdot \mathbb{1}[t = s]$ survive because they vary across **both** individuals and time.

Two-way fixed effects

- Combining individual and time fixed effects gives the **two-way fixed effects** (TWFE) model:

$$\begin{aligned} Y_{it} = {} & \alpha_i + \lambda_t \\ & + \sum_{s \neq 0} \beta_s \cdot D_i \cdot \mathbb{1}[t = s] + U_{it}, \end{aligned}$$

  where α_i is the individual fixed effect and λ_t is the time fixed effect.

- This is the standard specification for event studies and DID with multiple periods.